

YatraDrishti

Urban Fleet Intelligence Copilot

"AI-driven risk intelligence for every vehicle, every journey."

Team: **Zenbytes**

Track: **Open**

Hackathon: **IBM Bob AI Hackathon 2026**

Jemit Vaghasiya(TL) · Jay Sohaliya · Yug Bhensadadiya · Yug Bhungaliya

YatraDrishti

INTELLIGENCE FOR EVERY JOURNEY

Smarter Journeys for a Better Tomorrow

AI-powered fleet intelligence for safer, greener and more efficient roads.



Real-time Insights



Optimized Routes



Safer & Greener Journeys

On Track for a Greener Tomorrow



PEOPLE
PLANET
PROGRESS

Driven by Data.
Towards a Sustainable Tomorrow."

Every Journey Matters

Fleet managers are flying blind.

WHO HAS THIS PROBLEM?

- Fleet dispatchers managing 20–500 vehicles
- Safety officers tracking driver behaviour
- SME logistics operators on Indian highway corridors (NH-48, JNPT, Chennai Port)

THE CORE PAIN

- Risk signals buried in spreadsheets & siloed GPS tools
- Diagnosis is always reactive — after the delay/accident happened
- No AI early-warning, no plain-language query interface

WHY IT MATTERS — COST OF INACTION

₹20K

per hour demurrage (port corridors)

30%

excess fuel from unmonitored vehicles

3–4

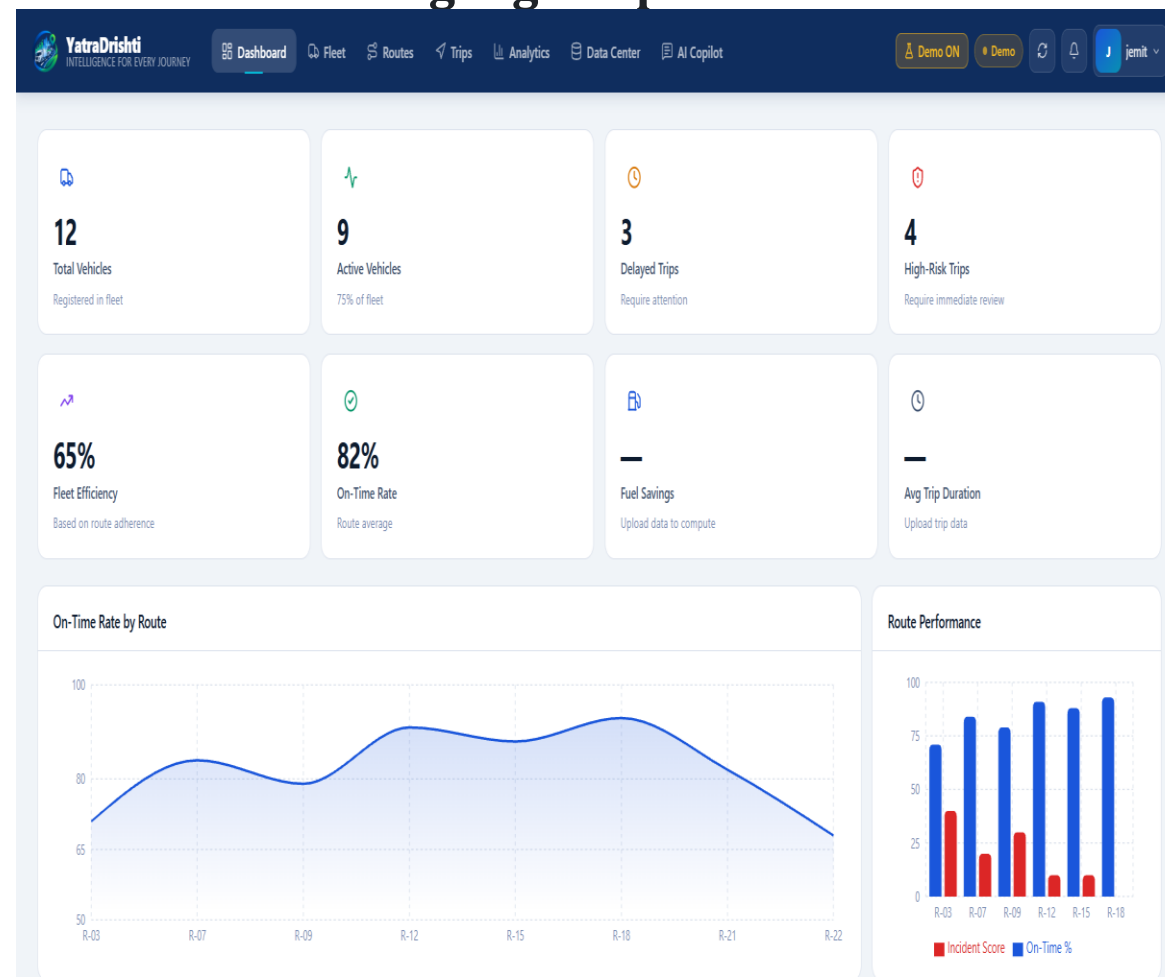
disconnected tools needed to diagnose one incident

 Buried signals + reactive diagnosis + no AI early-warning = costly, avoidable downtime.

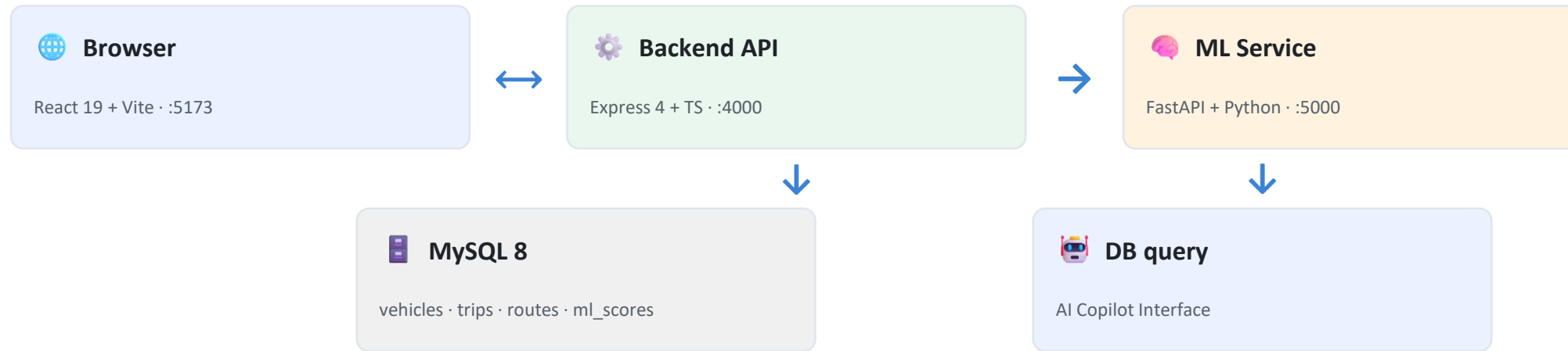
Upload one CSV → instant AI risk scores + live dashboard + natural-language Copilot.

KEY CAPABILITIES

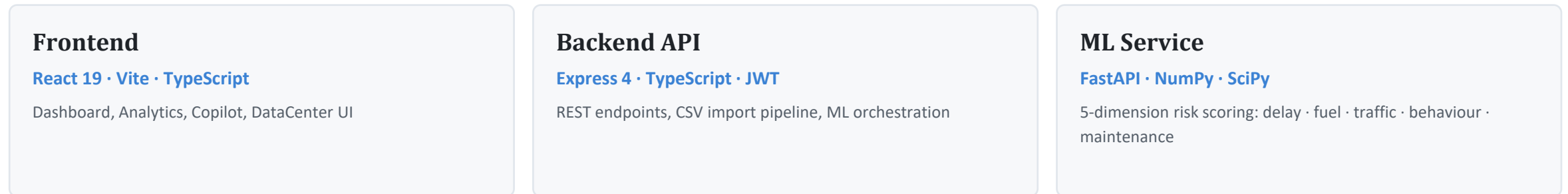
- ML Risk Scoring — every vehicle & trip scored LOW → CRITICAL
- Live Dashboard — KPI tiles, charts, fleet risk table
- IBM Bob Copilot — ask fleet questions in plain language
- CSV-first — works with any basic GPS export, no hardware needed
- Demo Mode — full Indian fleet dataset, no backend required



SYSTEM FLOW



COMPONENT BREAKDOWN



From raw CSV → ML risk scores → dashboard → AI answer in under 10 seconds.

STEP-BY-STEP FLOW

- Step 1 — Upload trips.csv on Data Center page
- Step 2 — Backend validates + stores in MySQL, calls ML service
- Step 3 — ML scores each trip & vehicle (LOW → CRITICAL)
- Step 4 — Dashboard & Analytics populate instantly
- Step 5 — Ask Copilot: "Which vehicles are at critical risk?"

hard-fleet (3).csv
2.7 KB · 50 data rows

Preview — 50 valid row(s)

VEHICLE_ID	ROUTE	DISTANCE_KM	EXPECTED_TIME	ACTUAL_TIME	FUEL_USED	TRAFFIC
VH-001	Downtown Core Loop	48.0	95	142	7.8	Standstill
VH-002	Northside Industrial	62.0	110	168	11.4	Heavy
VH-003	Airport Connector	35.0	70	107	6.2	Heavy
VH-004	South Port Freight	78.0	135	198	26.4	Standstill
VH-005	East Residential Circuit	55.0	100	151	10.9	Heavy
VH-006	West Cross-Town Express	42.0	85	134	8.7	Standstill
VH-007	Downtown Core Loop	48.0	95	128	7.1	Heavy
VH-008	Airport Connector	35.0	70	118	7.8	Heavy
VH-009	Northside Industrial	62.0	110	179	13.6	Standstill
VH-010	South Port Freight	78.0	135	215	28.9	Standstill
VH-011	East Residential Circuit	55.0	100	162	12.4	Heavy
VH-012	West Cross-Town Express	42.0	85	140	9.3	Standstill
VH-001	South Port Freight	78.0	135	191	24.8	Standstill

Which drivers need safety coaching?
You 07:40 PM

Driver Safety Attention Needed:

- Driver on **TRK-5512** (Carlos Rivera): vehicle at CRITICAL risk, speed: 0 mph
- Driver on **TRK-4456** (Michael Torres): vehicle at CRITICAL risk, speed: 61 mph

Recommendation: Schedule safety coaching sessions for drivers operating high-risk vehicles.

07:40 PM

Ask Bob about your fleet data...

6 IBM Technologies — How We Used Them

PRIMARY IBM TECHNOLOGY

IBM Bob

AI Copilot Platform · Used 2 ways

1. Development environment: The full project was built inside IBM Bob — agent mode, skill activation, code generation, documentation, and architecture planning were all done using Bob's workspace throughout the hackathon.

WHAT IBM BOB ENABLED

- Instant code generation & refactoring across TypeScript, Python, and CSS in one workspace
- Skill-driven documentation — architecture, problem statement, setup guide all authored with Bob



Hi, I'm Bob

Ask me questions or let me code for you.

 Documentation

A fleet manager goes from raw spreadsheet to AI-graded risk intelligence in one upload.

QUANTIFIED OUTCOMES

<10s

CSV upload to full dashboard

5

risk dimensions scored per entity

4

risk levels:
LOW·MED·HIGH·CRITICAL

0

hardware or integrations needed

BUSINESS IMPACT

- Early risk flag on a single critical vehicle can avoid ₹20K/hr demurrage on port corridors
- Per-vehicle fuel scoring identifies 20–30% fuel outliers invisible to basic GPS tools
- Copilot reduces manual diagnosis from 45 min → under 1 min per incident

WHAT A JUDGE SEES IN THE DEMO

- ✓ Upload sample CSV → scores appear instantly
- ✓ Dashboard KPIs populated from real imported data
- ✓ Analytics charts show delay + fuel from that CSV
- ✓ Copilot answers a natural-language fleet query
- ✓ Demo Mode works fully offline — no DB needed



Zenbytes · CHARUSAT University

Track: Open · IBM Bob AI Hackathon 2026

Jemit Vaghasiya

Team Lead · Full-Stack

Architecture design, backend API (Express + MySQL), ML scoring pipeline, project coordination

Jay Sohaliya

Frontend Developer

React frontend, Dashboard & Analytics pages, Recharts visualisations, Demo Mode

Yug Bhensadadiya

ML Engineer

FastAPI ML microservice, 5-dimension risk scoring model (NumPy / SciPy), batch scoring endpoints

Yug Bhungaliya

UI / Data

Data Center page, CSV import flow, Auth UI, mock data (Indian fleet), documentation

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Thank You